

GASTROPODS

Gastropods are by far the largest class of molluscs. The word gastropod means “stomach-foot”, because the animal appears to crawl on its belly.

Most snails and slugs glide along a stream of slime on a single muscular foot, rasping at their food with a radula, a tongue-like emery-paper. Most gastropods use their radula to rasp at vegetation, algae, or the fine film of microbes that coat underwater rocks, but some are predatory. They usually have a distinct head with well-developed sensory tentacles. Snails have a coiled or helical shell into which they can withdraw; slugs lost their shell during the course of evolution. The class also includes a number of animals that deviate from this basic plan, such as swimming

sea slugs. Gastropods originated in the oceans and they still have their greatest diversity there; other species live in fresh water or on land.

TORSION

The juvenile gastropod undergoes a process called torsion: the whole of the body within the shell twists round 180 degrees, so that the respiratory mantle cavity comes to lie above the animal's head. This allows the vulnerable head to withdraw into the safety of the shell. In marine snails, such as periwinkles and limpets, this body form persists into adulthood. Because of this, marine snails are described as prosobranchs, meaning “forward gills”. In sea slugs, however, the body twists back again, and they are called opisthobranchs, or “hind gills”.

PHYLUM	MOLLUSCA
CLASS	GASTROPODA
ORDERS	21
FAMILIES	409
SPECIES	About 67,000

DEBATE

A COMMON TWIST

Sea snails were originally grouped together because their body twists during development so that their gills face forwards. But the ancestor of all snails probably had this feature, and now marine snails are split into groups based on other characteristics. Their exact relationships remain debatable.

TRUE LIMPETS

Primitive algae-grazing limpets, the Patellogastropoda have a slightly coiled, conical shell. Their powerful muscles clamp down firmly on intertidal rocks, protecting them from predators, desiccation, and waves.

COMMON LIMPET

Patella vulgata
F: Patellidae

This species grazes algae from rocks on shores of the N.E. Atlantic at high tide, then returns to a depression in the rock that matches the shape of its shell.



NERITES AND RELATIVES

A small but diverse group, Cycloneritimorpha are well known in the fossil record. They include marine, freshwater, and land-living forms, some with a coiled shell, and a few like limpets. There are also species with a trapdoor to their shell.

ZIG-ZAG NERITE

Neritina communis
F: Neritidae

An inhabitant of Indo-Pacific mangroves, this is a highly variable species, with white, black, red, or yellow shells even in the same population.



BLEEDING TOOTH NERITE

Nerita peloronta
F: Neritidae

Named for the blood-red mark on its shell opening, this Caribbean intertidal mollusc can survive out of water for prolonged periods.

TOP SHELLS AND RELATIVES

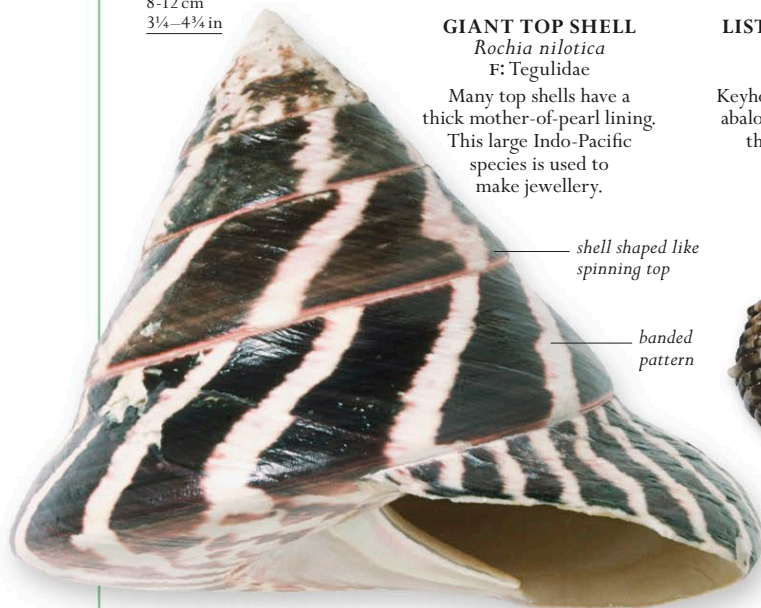
The Vetigastropoda are marine gastropods that graze algae and microbes with a brush-like radula. The shells range from those of keyhole limpets, with the barest suggestion of a spire, to the coiled globes and pyramids of top shells, which can seal themselves inside using an operculum, or trapdoor.

8-12 cm
3 1/4-4 1/4 in

GIANT TOP SHELL

Rochia nilotica
F: Tegulidae

Many top shells have a thick mother-of-pearl lining. This large Indo-Pacific species is used to make jewellery.



LISTER'S KEYHOLE LIMPET

Diodora listeri
F: Fissurellidae

Keyhole limpets are close relatives of abalones and top shells. Like others, this W. Atlantic species ejects oxygen-depleted water through its “keyhole”.



SILVER MOUTH TURBAN SHELL

Turbo argyrostomus
F: Turbinidae

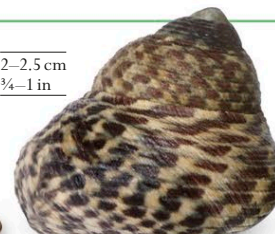
Turban shells are close relatives of top shells, but have a calcified operculum. This is an Indo-Pacific species.

2-2.5 cm
3/4-1 in

CHECKERED TOP SHELL

Phorcus turbinatus
F: Trochidae

Top shells have a “spinning-top” shell that is closed with a circular operculum. This is a Mediterranean species.



RED ABALONE

Haliotis rufescens
F: Haliotidae

Abalones have an ear-shaped shell lined with thick mother-of-pearl and punctuated with holes for exhaling water. This N.E. Pacific kelp-grazing species is the largest.